

ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION TECHNOLOGY MODULE 2
UNIT NAME: COMPUTER REPAIR AND MAINTAINANCE
UNIT CODE: 0714 351 04A
PRACTICAL 1, TERM 2
LEVEL 5
TOTAL MARKS 50

Instructions to the Student

- ✓ *Read the whole paper carefully before you begin.*
- ✓ *Follow all safety rules. **Unsafe practice may lead to the assessor stopping the test.***
- ✓ *Work on your own. Ask the assessor only if there is a safety concern or a missing item.*
- ✓ *Record each stage on the job card as you work.*
- ✓ *Hand in your job card and leave your workstation clean and tidy.*
- ✓ *Do not open the power supply unit (PSU) at any time.*

1. Scenario

A client has brought a desktop computer to your workshop with the following complaints:

1. The computer becomes very hot and the fans are loud. It shuts down or slows down after a short time.
2. The computer sometimes shows a **Blue Screen of Death (BSOD)**.

3. At other times the screen displays the error message "**No bootable device**".
4. The monitor screen and casing are dusty and dirty.

Initial inspection shows heavy **dust accumulation** on the CPU fan, heatsink, and case fans. The component causing the blue screen and boot errors has **not been identified**, and you must find it yourself. A tray of spare parts has been provided.

Your task: Diagnose the faults, identify the faulty component, clean the system, repair and replace the faulty part using the spare parts provided, reassemble and test the computer, and dispose of the faulty part correctly. Document your work on the job card at the end of this paper.

2. Materials and Equipment Provided

Hardware

- 1 desktop computer (faulty), with keyboard, mouse, monitor, and power cable
- Spare parts tray: 1 HDD (tested and working, with the operating system already installed, and SATA data and power cables) and other spare parts
- Thermal paste (small tube or sachet)
- Cable ties (zip ties) and Velcro straps

Cleaning materials

- **Air blower** (hand blower or compressed air can) for removing dust
- **Microfibre cloth** for cleaning the monitor screen and casing
- Soft anti-static brush
- Isopropyl alcohol (90% or higher) and lint-free swabs/cotton buds
- Dust mask and safety goggles

Tools

- Screwdriver set (Phillips #1 and #2, flathead, and Torx if needed)
- Anti-static wrist strap and anti-static mat
- Anti-static bags (for the removed part)
- Pliers (needle-nose) and tweezers
- Plastic spudger/opening tool
- Magnetic parts tray or small labelled containers for screws
- Flashlight or work lamp
- Marker pen and masking tape (for labelling)

Documentation

- Motherboard/computer instruction (service) manual
- Job card/repair log (Section 6 of this paper)
- Pen, ruler, and notepad

3. Safety Requirements

Before starting, you must:

- Wear an **anti-static wrist strap** clipped to the bare metal of the case.
- Wear a **dust mask and safety goggles** when blowing out dust.
- Shut down and **unplug the power cable** before opening the case.
- Work on a clean, flat, well-lit surface away from liquids.
- Blow dust **outdoors or in a ventilated area**, away from other workstations.

4. Tasks

Task 1: Preparation and Initial Inspection

- 1.1 Put on your safety equipment and prepare your workstation.
- 1.2 Connect the computer, power it on, and **observe and record the symptoms**: fan noise, heat, any BSOD, any boot error messages, and any beep codes. Then shut it down and unplug it.
- 1.3 Refer to the manual, and list the tools you will use.

Task 2: Disassembly and Diagnosis

- 2.1 Disconnect all external cables and peripherals.
- 2.2 Remove the side panel and take **photographs or draw a sketch** of the internal layout and cable connections.
- 2.3 **Identify the faulty component** using the symptoms you recorded, the BIOS/UEFI screen, a cable and connection check, and the diagnostic tools provided. Write your reasoning on the job card. Then disconnect and remove the faulty part, place it in an **anti-static bag**, and label it "FAULTY".
- 2.4 Remove or loosen the CPU cooler and case fans only as far as needed for cleaning, following the manual. Keep all screws in a labelled tray.

Task 3: Cleaning and Dust Removal

- 3.1 Using the **air blower**, remove dust from the following, holding the fan blades still with a finger or swab so they do not over-spin:

- CPU fan and heatsink fins
- Case fans (front and rear)
- PSU vents (outside only)
- Motherboard, RAM slots, and expansion card slots

- Vents and drive bays

3.2 Use a soft brush and swabs lightly dampened with isopropyl alcohol for stubborn dust.

3.3 Check the CPU heatsink. If the thermal paste is dried out, **clean the old paste and apply fresh thermal paste** (pea-sized amount) before refitting the cooler.

3.4 Use the **microfibre cloth** to clean the monitor screen and the outside of the casing. Use gentle strokes with no liquid sprayed directly on the screen.

Task 4: Installing the Replacement Part

4.1 Select the correct replacement from the spare parts tray, check that it is **compatible** (interface, size, and connectors), and record its details (make, capacity, interface) on the job card.

4.2 Mount the replacement part in its correct position and secure it with screws.

4.3 Connect all required data and power cables correctly and firmly.

4.4 Check that all cables are tied neatly and do not block any fan.

Task 5: Reassembly and Testing

5.1 Refit the CPU cooler and fans (if removed), confirm that the fan power connectors are plugged in, and close the case.

5.2 Reconnect the peripherals and power cable, then power on. Confirm that the computer passes **POST** with no error beeps.

5.3 Enter the **BIOS/UEFI setup** and confirm that all components are detected. Check the boot order, and set the correct boot device first.

5.4 Verify that the repair worked:

- The system boots to the desktop with **no "No bootable device" message**

- No BSOD appears after about 10 minutes of use
- Fans run normally and the temperature (BIOS or monitoring software) is in the normal range
- The screen is clean and clear

Task 6: Disposal and Final Clean-up

6.1 State, on the job card, how the **faulty component you removed** should be disposed of properly. Mention **data sanitization** (wiping, degaussing, or physical destruction) if it stores data, and a **certified e-waste recycler or take-back programme**.

6.2 Name two environmental or health hazards of dumping electronic waste in a landfill or burning it.

6.3 Place the faulty component in the designated e-waste collection box. Do not throw it in the ordinary bin.

6.4 Return all tools, put away all materials, and leave the workstation clean.

5 Job Card (to be completed by the student)

Item	Details
Computer make/model	
Date received	
Client complaint	
Symptoms observed	
Diagnosis steps and reasoning	
Faulty component identified	
Cause of overheating	
Cause of BSOD and "No bootable device" error	
Tools and materials used	
Cleaning done	
Part replaced (make, capacity, interface)	
Test results after repair	
Disposal of faulty part	
Student signature	
Assessor signature	

The trainer is introducing oral questions shortly